Universidade do Minho
Departamento de Informática

Dados e Aprendizagem Automática
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Practical Exercise no. 8

Theme

Artificial Neural Networks

Exercise

Artificial Neural Networks (ANNs) are computer models that mimic the human central nervous system. They are capable of solving classification and regression problems.

A MultiLayer Perceptron (MLP) is a fully connected class of feedforward ANN. It contains at least three layers of nodes, and, except for the input nodes, each node is a neuron that typically uses a non-linear activation function. MLPs form the basis of neural networks and can improve computational performance when applied to classification or regression problems.

Tasks

This practical exercise includes the following tasks:

T1. Follow and implement the steps presented in the practical notebooks to design and optimize neural networks using the datasets from the practical group work. Use PyTorch for classification problems.

T2. Optimize the previous networks by varying additional hyperparameters, such as the number of neurons or layers in the network. How do these changes affect the network's performance? What is the best network configuration?

T2. Vary the training hyperparameters to improve the results obtained by the previous networks. Adjust the number of epochs, the learning rate, the loss function and the optimization function. What is the optimal set of hyperparameters and what accuracy does this correspond to?

T3. How does performance vary when compared to models such as Decision Trees or Random Forests?